

ENR 1 GENERAL RULES AND PROCEDURES

ENR 1.1 GENERAL RULES

1. General Rules and Procedures

- 1.1 The air traffic rules and procedures applicable to air traffic within the Manila FIR conform to Annex 2 and Annex 11; to the Convention on International Civil Aviation; to the Civil Aeronautics Act of the Philippines (R.A. No. 776) and to the appropriate national rules and procedures listed in GEN 1.6; to those portions of the Procedures for Air Navigation Services – Air Traffic Management applicable to aircraft and of the Regional Supplementary Procedures applicable to the South East Asian and Pacific Region except for the differences listed in GEN.
- 1.2 With the exception of military and private aerodromes, air traffic services (ATS) in the Philippines are provided by the CAA of the Philippines (CAAP), administered by the Assistant Director General, Air Traffic Service.
- 1.3 The airspace of the Philippine territory, including adjacent international waters, comprise a single FIR (Manila FIR) where ATC is exercised:
- a. on airways covering the main ATS routes; and
 - b. in terminal control areas and control zones at controlled aerodromes equipped with approach and landing lights.
- 1.4 Flight information and alerting services within the FIR and ATC service in control areas are provided by Manila ACC within its areas of responsibility.
- 1.5 A maximum of 1500 FT/minute rate of climb/descent shall be observed by pilots prior to reaching the last 2000 FT of its assigned altitude/level.

2. Dropping of Objects

- 2.1 Nothing shall be dropped or sprayed from an aircraft in flight except under conditions prescribed by the Director General of Civil Aviation (DGCA), and as indicated by relevant information, advice and/or clearance from the appropriate ATS unit.

3. Acrobatic Flying

- 3.1 No aircraft shall be flown acrobatically except under conditions prescribed by the DGCA, and as indicated by relevant information, advice and/or clearance from the appropriate ATS unit.

4. Towing Flights

- 4.1 No aircraft or other object shall be towed by an aircraft except in accordance with the requirements prescribed by the DGCA, and as indicated by relevant information, advice and/or clearance from the appropriate ATS unit.

5. Aerial Photography

- 5.1 Aerial photography of Philippine territory is prohibited without prior authorization from the Philippine Air Force, Villamor (Nichols) Air Base, Pasay City.

6. Times and Units of Measurement

- 6.1 Coordinated Universal Time (UTC) and the prescribed units of measurement shall be applied to flight operations.

7. Airspace Structure

- 7.1 For the performance of the flight information service and the alerting service, CAAP establishes flight information regions which are published in the AIP. Within the FIR, CAAP establishes the controlled and uncontrolled airspace according to the extent of the ATS maintained thereof, based on the classification described in subsection ENR 1.4. Within controlled airspace, VFR flights may be prohibited completely or partly by the ATS with regard to limitation of space and time if urgently required by the degree of intensity of air traffic subject to ATC.

8. Prohibited, Restricted and Danger Areas

- 8.1 Within the Manila FIR, several prohibited, danger and restricted areas have been established. Particulars are given in ENR 5.1.
- 8.2 Activation of areas subject to intermittent activity is notified well in advance by a NOTAM, giving reference only to the designation of the areas.

9. Parachute Descents

- 9.1 Parachute descents, other than emergency descents, shall not be made except under conditions prescribed by the DGCA, and as indicated by relevant information and/or clearance from the appropriate ATS unit.

10. Flight Training Areas

- 10.1 Several training areas have been established within Manila FIR. The particulars of these are given in ENR 5.2.

11. Communications Failure

- 11.1 Air Ground
- 11.1.1 When an aircraft fails to establish contact with the aeronautical station on the designated frequency, it shall attempt to establish contact on another frequency appropriate to that route. If this attempt fails, the aircraft shall attempt to establish communication with other aircraft or other aeronautical stations, including ocean vessels, where applicable, on frequencies appropriate to that route. In addition aircraft operating within a network shall monitor the appropriate VHF frequency for calls from nearby aircraft.
- 11.1.2 If the attempts specified above failed, the aircraft station shall transmit its message twice on the designated frequencies, preceded by the phrase "TRANSMITTING BLIND" and, if necessary, including the addressee(s) for which the message is intended.
- 11.1.3 Aircraft forming part of the aerodrome traffic at a controlled aerodrome can provide notice of a requirement to land by applying the procedures below:
- a. observe and sequence itself from other traffic prior to entering the normal traffic circuit of intended runway-in-use and watch for light signals from the Control Tower. If no light signal is received, aircraft shall fly at an appropriate altitude and when abeam the Control Tower, light aircraft shall rock its wings and heavier aircraft shall flash on and off twice its landing or navigation lights;
 - b. if still no light signal is received and the intended runway-in-use is clear of aircraft or vehicle it shall continue its approach and land; and
 - c. contact the Control Tower by telephone (or any other means of communications) immediately after landing.

- 11.1.4 If in VISUAL METEOROLOGICAL CONDITIONS (VMC), the aircraft shall:
- continue to fly in visual meteorological conditions, and
 - land at the nearest suitable aerodrome; and
 - report its arrival by the most expeditious means to the appropriate air traffic control unit.
- 11.1.5 If in INSTRUMENT METEOROLOGICAL CONDITIONS (IMC) or when weather conditions are such that it does not appear feasible to complete the flight in accordance with 11.1.4 above, the aircraft shall:
- proceed according to the current flight plan to the appropriate designated navigational aid serving the aerodrome of intended landing and, when required to ensure compliance with item b) below, hold over this aid until commencement of descent;
 - commence descent from the navigational aid specified in item a) above at, or as close as possible to, the expected approach time last received and acknowledged; or, if no expected approach time has been received and acknowledged, at, or as close as possible to, the estimated time of arrival as indicated in the filed flight plan and revised in accordance with the current flight plan;
 - complete a normal instrument approach procedure as specified for the designated navigational aid; and
 - land, if possible, within 30 minutes after the estimated time of arrival specified in item b) above or the last acknowledged expected approach time, whichever is later.

Note:

- As evidenced by meteorological conditions prescribed therein, 11.1.4 relates to all controlled flights, whereas 11.1.5. relates only to IFR flights.
- If the clearance for the levels covers only part of the route, the aircraft is expected to maintain the last assigned and acknowledged cruising level(s) to the point(s) specified in the clearance and thereafter the cruising level(s) in the current flight plan.

11.2 Aircraft Receiver Failure

- 11.2.1 When an aircraft station is unable to establish communication due to receiver failure, it shall transmit reports at the scheduled times, or positions, on the frequency in use, preceded by the phrase "TRANSMITTING BLIND DUE TO RECEIVER FAILURE". The aircraft station shall transmit the intended message, following this by a complete repetition. During this procedure, the aircraft shall also advise the time of its next intended transmission.
- 11.2.2 An aircraft, which is provided with air traffic control or advisory services, shall, in addition to complying with 11.2.1, transmit information regarding the intention of the pilot-in-command with respect to the continuation of the flight of the aircraft.

12. Automatic Dependent Surveillance - Contract/Controller - Pilot Data Link Communication (ADS-C/CPDLC) Implementation within Manila FIR

12.1 Introduction

12.1.1 Data link services are available to FANS 1/A capable aircraft in the following air routes in the Manila FIR:

A. East Sector

1. A582

2. A590

3. G578

4. M501

5. G467

B. North Sector

1. A583

C. South Sector

1. P515

2. R337

3. A339

4. B462

5. B473

D. West Sector

1. N892

2. L625

3. L628

4. M772

5. M765

6. W17

12.1.2 The implementation of ADS-C/CPDLC does not affect the current procedures for non-data link capable aircraft.

12.2 Data Link Airspace and Means of Communications

SECTOR	DATA LINK AIRSPACE	COMMUNICATION	
		PRIMARY	SECONDARY
EAST	A582	CPDLC	HF
	A590		
	G578		
	M501		
	G467		
NORTH	A583		1. VHF 2. HF
SOUTH	P515		HF
	R337		
	A339		
	B462		
	B473		
WEST	N892		HF
	L625		
	L628		
	M772		
	M765		
	W17		

12.3 Log On Procedures

- 12.3.1 The log on is the first step in the data link process. A log on, initiated either by the flight crew or by another ATS Unit (ATSU), is performed prior to the ATSU establishing a CPDLC and/or ADS-C connection.
- 12.3.2 An initial log on request is needed when the aircraft does not have an ADS-C or CPDLC connection.
- 12.3.2.1 The ATS Facilities Notification (AFN) log on is performed by the pilot when:
- the aircraft is preparing for departure; or
 - the aircraft will enter an area, where data link services are available, from an area where data link services are not available; or
 - instructed by ATC (e.g. following a failed data link transfer).
- 12.3.2.2 The AFN log on address of Manila ACC is "RPHI".
- 12.3.2.3 Aircraft should log on to "RPHI" prior to departing from Ninoy Aquino International Airport (NAIA) or between 10 - 25 minutes prior to entering the Manila FIR.
- 12.3.2.4 To avoid an automatic rejection of the log on request, the flight crew shall ensure that the flight-specific information entered into the aircraft system is the same as the corresponding details filed in the flight plan.
- 12.3.2.5 Pilots who are unable to establish a data link connection shall inform ATC on VHF or HF.

12.4 ADS/CPDLC Procedures

- 12.4.1 Aircraft with ADS/CPDLC connection with RPHI shall send ADS position report upon entering the FIR boundary.

- 12.4.2 ADS-C reports fulfill all normal position reporting requirements.
- 12.4.3 In airspace where both VHF voice and CPDLC services are available, the flight crew should use VHF as means of communication.
- 12.4.4 The response to a CPDLC message should be via CPDLC, and a response to voice should be via voice.
- 12.4.5 Exchange of CPDLC Messages
 - 12.4.5.1 The ATS system will receive a network acknowledgement (MAS Message Assurance) to an uplink message indicating that the message has been delivered to the aircraft's ACARS Management Unit (MU); and
 - 12.4.5.2 The avionics will receive a network acknowledgement to a downlink message indicating that the message has been delivered to the communication service provider's system.
 - 12.4.5.3 A controller or pilot having any doubt as to the intent of a message, or if any other ambiguity exists, clarification shall be sought through the use of voice communication.
 - 12.4.5.4 Read back for ATC clearances/instructions issued via CPDLC is not required.
- 12.4.6 Uplink/Downlink Messages
 - 12.4.6.1 The WILCO downlink message indicates that the pilot will comply fully with the clearance/instruction contained in the associated uplink message.
 - 12.4.6.2 The ROGER or AFFIRM uplinks are not appropriate responses and shall not be used as a response to a clearance request. The controller shall approve a clearance request by sending an uplink message containing the actual clearance.
 - 12.4.6.3 AFFIRM is an appropriate response to an uplinked negotiation request message that is acceptable (e.g. CAN YOU ACCEPT [altitude] AT [time]).
 - 12.4.6.4 UNABLE will be used as a negative response to a clearance request. It also applies as a negative response to all elements of a multi-element request. The clearance shall not be restated.
 - 12.4.6.5 To avoid ambiguity in message handling and response, a CPDLC downlink message should not contain more than one clearance request.
- 12.4.7 Use of Pre-Formatted and Free Text Messages
 - 12.4.7.1 Creation of a clearance request and the issuing of a clearance shall be performed by the use of pre-formatted message elements.
 - 12.4.7.2 Free text messages shall be used only when an appropriate pre-formatted message element does not exist.
 - 12.4.7.3 When a free text message is required, standard ICAO phraseology, abbreviations and format shall be used.
- 12.4.8 CPDLC Connection Transfer and End of Service
 - 12.4.8.1 CPDLC Connection Transfer from RPHI to Next Data Authority as appropriate.
 - 12.4.8.1.1 RPHI shall terminate the CPDLC connection by:
 - a. an uplink message of "MONITOR (OR CONTACT) [icaounitname] [frequency] 10 minutes before the FIR boundary, and;

- b. an uplink message of “END SERVICE” after receiving WILCO from the pilot.

12.4.8.1.2 The avionics will downlink a “DISCONNECT” message.

12.4.8.1.3 In cases where the next FIR provide data link services; a Next Data Authority (NDA) message will be sent out 30 - 35 minutes prior to crossing the FIR boundary. Transfer of communication shall be completed at the FIR boundary.

12.4.8.1.4 In cases where the next FIR does not have data link services, CPDLC connections will be terminated at the FIR boundary position.

12.5 Flight Plan Procedures

12.5.1 In Item 10 of the flight plan, the operator should insert one or more of the descriptors, as appropriate, listed in Table 1.1, to identify an aircraft’s data link equipment and capabilities:

Table 1.1: Descriptors for CPDLC/ADS-C equipment and capabilities in Item 10

Item 10a - CPDLC equipment and capabilities	Descriptor
CPDLC ATN VDL Mode 2 (ATN B1)	J1
CPDLC FANS 1/A HFDL	J2
CPDLC FANS 1/A VDL Mode 0/A	J3
CPDLC FANS 1/A VDL Mode 2	J4
CPDLC FANS 1/A SATCOM (INMARSAT)	J5
CPDLC FANS 1/A SATCOM (MTSAT)	J6
CPDLC FANS 1/A SATCOM (Iridium)	J7
RCP 400	P1
RCP 240	P2
Item 10b - ADS-C equipment and capabilities	Descriptor
ADS-C with FANS 1/A capabilities	D1
ADS-C with ATN capabilities	G1

12.5.2 In Item 18 of the flight plan, the operator should insert the following other information relevant to CPDLC and ADS-C equipment and capabilities:

- a. The indicator REG/ followed by the aircraft registration;
- b. The indicator CODE/ followed by the aircraft address expressed in the form of an alphanumeric code of six hexadecimal characters; and
- c. The indicator SUR/ followed by RSP400 or RSP180, as appropriate.

12.6 Emergency Procedures

12.6.1 The flight crew shall use whatever means appropriate (i.e. CPDLC and/or voice) to communicate during an emergency.

12.6.2 When emergency situations are communicated via CPDLC, the controller may respond via CPDLC. However, the controller may also attempt to make voice contact with the aircraft.

12.6.3 If the ATSU receives a CPDLC emergency message such as DM 55 PAN PAN PAN or DM 56 MAYDAY MAYDAY MAYDAY, the controller should acknowledge receipt of the CPDLC message using the most appropriate means (voice or CPDLC). If responding by CPDLC, the controller should use either of the following free text message elements (as appropriate):

- a. UM 169r ROGER PAN if the downlink message contains DM 55 PAN PAN PAN; or

- b. UM 169q ROGER MAYDAY if the downlink message contains DM 56 MAYDAY MAYDAY MAYDAY.

Note 1: For Future Air Navigation System (FANS) 1/A, the CPDLC emergency messages do not require a closure response. Therefore, the aircraft system will reject receipt of any technical response (i.e. including an MRN), such as the DM 3 ROGER message element.

Note 2: For FANS 1/A, if the controller sends a CPDLC free text messages to respond to an emergency message, the flight crew may not send the required response (i.e. DM 3 ROGER) to the free text message, depending on workload and the nature of the emergency.

- 12.6.4 If the emergency situation no longer exists, the pilot should cancel the ADS emergency mode (if activated).

12.7 Data Link Service Failure

12.7.1 CPDLC Connection Failure

- 12.7.1.1 If a CPDLC dialogue is interrupted by a data link service failure, the controller should recommence the entire dialogue by voice communication.

- 12.7.1.2 When the pilot recognizes a failure of the CPDLC connection, the controller should instruct the flight crew to terminate the connection and then initiate another log on. The controller or radio operator should use the following voice phraseology:

Controller (or Radio Operator)	CPDLC FAILURE. DISCONNECT CPDLC THEN LOG ON TO [facility designator]
Flight crew	ROGER

12.7.2 Data Link Service Failure

12.7.2.1 Unplanned Data Link Shutdown

- 12.7.2.1.1 In the event of an unplanned data link shutdown, the relevant ATSU should inform:

- a. All affected aircraft using the following voice phraseology:

Controller (or Radio Operator)	ALL STATIONS CPDLC FAILURE. DISCONNECT CPDLC. CONTINUE ON VOICE
Flight crew	ROGER

- b. The adjacent ATSUs by direct coordination; and

- c. All relevant parties via the publication of a NOTAM; if appropriate.

12.7.2.2 Planned Data Link Shutdown

- 12.7.2.2.1 During the time period of a planned data link shutdown, the ANSP will advise the operators of the requirements to use voice communication procedures.

- 12.7.2.2.2 When advising the flight crew prior to the commencement of the planned data link shutdown, the controller should use the following CPDLC message or the radio operator should use the equivalent voice phraseology:

Controller (or Radio Operator)	CPDLC WILL BE SHUT DOWN. DISCONNECT CPDLC. CONTINUE ON VOICE. Note 1: The controller could operationally provide the voice frequency.
Flight crew	ROGER Note 2: The controller expects the flight crew to terminate CPDLC.

12.7.3 Resumption of Data Link Operations

- 12.7.3.1 The controller or radio operator should use the following voice phraseology to advise the flight crew that the CPDLC and ADS-C system has resumed operations:

Controller (or Radio Operator)	[ALL STATIONS] RESUME NORMAL CPDLC and ADS-C OPERATIONS.
Flight crew	LOG ON [facility designation]

13. Performance-Based Communication and Surveillance (PBCS) Implementation within the Manila FIR

13.1 Introduction

- 13.1.1 Civil Aviation Authority of the Philippines (CAAP) plans to implement Performance-Based Communication and Surveillance (PBCS) in accordance with the ICAO provisions to support the Performance-Based reduced horizontal separation minima application using data link.
- 13.1.2 PBCS is a concept that enables the management of communication and surveillance capabilities by prescription of Required Communication Performance (RCP) and Required Surveillance Performance (RSP) specifications in Future Air Navigation System (FANS 1/A) data link operations using the Automatic Dependent Surveillance - Contract (ADS-C) and Controller - Pilot Data Link Communications (CPDLC).

13.2 PBCS Framework

- 13.2.1 The PBCS concept provides a framework to apply RCP and RSP specifications to ensure the acceptable communication and surveillance capabilities and performance of an operational system.
- 13.2.2 The main components that involve the joint participation from States, ANSPs and aircraft operators under the PBCS implementation framework consist of the following:
- To prescribe RCP and RSP specifications, for aircraft operators, aircraft systems and infrastructure supporting data link operations, when applying separations predicted on such performance;
 - Operational approval of aircraft operators for a communication and/or surveillance capability including aircraft equipment for operations where RCP and/or RSP specifications will have to be prescribed;
 - Indications of an aircraft's communication and performance capability in the form of RCP/ RSP specifications in the flight plan, and;

- d. Monitoring programmes to assess actual communication and surveillance performance against RCP and RSP specifications and to determine corrective action to report, analyze and resolve problems.

13.3 Operator and Aircraft Systems

- 13.3.1 Operator should continue to use CPDLC and ADS-C in accordance with policies established by the State of Registry or State of the Operator.
- 13.3.2 Operator obtaining approving in accordance with National regulations (State of the Operator or State of Registry) should ensure that the procedures, system and services in operations and maintenance programs meet the allocated criteria in interoperability standards and RCP/RSP specifications.
- 13.3.3 Establish the necessary training and qualification program for flight crews and flight operations officers/dispatchers in preparation for PBCS implementation that is consistent with ICAO Annex 1 and Annex 6.
- 13.3.4 Initial compliance to the PBCS requirements would be beneficial to establish the confidence that Air Traffic Management (ATM) operations could be provided only to eligible aircraft. The PBCS concept will enhance the safety on Performance-Based reduced horizontal separation minima application using data link Air Traffic Service (ATS) operations.

13.4 Separation Minima

- 13.4.1 Performance-Based Longitudinal Separation minima of 50 NM may be applied between RNP 10 and or RNP 4 approved aircraft and 30 NM may be applied between RNP 4 approved aircraft on designated air routes which either LOG ON to CPDLC or are within VHF radio range as the primary means of communication. Such Direct Controller-Pilot Communication (DCPC) shall be maintained at all times when applying these separation minima.
- 13.4.2 RCP240 and RSP180 performance specifications shall be required for the application of the Performance-Based Longitudinal Separation Minima in accordance with PANS-ATM (Doc 4444) paragraph 5.4.2.9.2.
- 13.4.3 Otherwise, 80 NM RNAV or 10 minutes (or less) Mach Number Technique (MNT) separation minima may be applied between aircraft in situation where DCPC could not be maintained or when RCP240/RSP180 performance requirement could not be complied.

13.5 Requirements

- 13.5.1 Operators conducting flights in air routes where separators are dependent on PBCS should start using RCP/RSP indicators in the flight plan beginning 23 May 2019 and to adhere to the provisions stated in Appendix 2 of PANS-ATM (Doc 4444).
- 13.5.2 Guidance material on the application of performance-based communication and performance-based surveillance, which prescribes RCP/RSP to an air traffic service in a specific area, is contained in the Performance-Based Communication and Surveillance (PBCS) Manual (Doc 9869).